

BEHAVIORAL INEQUALITY: THE CONTRIBUTION OF DECISION-MAKING FRICTIONS TO INEQUALITY

Stefano DellaVigna
UC Berkeley & NBER

Tim de Silva
Stanford GSB

Rohan Jha
Harvard

August 2026

Preliminary and Incomplete 😊

WHAT WE DO

- Individuals make many **choices** that are hard to justify with neoclassical models
 - Suboptimal refinancing, not claiming benefits, dominated insurance, ...

- Individuals make many **choices** that are hard to justify with neoclassical models
 - Suboptimal refinancing, not claiming benefits, dominated insurance, ...
- Our questions:
 - ① What are the **financial impacts** of these “frictions”, individually and in aggregate?
 - ② How much do they contribute to **inequality** in income, consumption, and wealth?

- Individuals make many **choices** that are hard to justify with neoclassical models
 - Suboptimal refinancing, not claiming benefits, dominated insurance, ...
- Our questions:
 - ① What are the **financial impacts** of these “frictions”, individually and in aggregate?
 - ② How much do they contribute to **inequality** in income, consumption, and wealth?
- Approach #1: **Empirical framework** + existing literature + existing data
 - ⇒ Estimates of losses across income distribution that are comparable across frictions

- Individuals make many **choices** that are hard to justify with neoclassical models
 - Suboptimal refinancing, not claiming benefits, dominated insurance, ...
- Our questions:
 - ① What are the **financial impacts** of these “frictions”, individually and in aggregate?
 - ② How much do they contribute to **inequality** in income, consumption, and wealth?
- Approach #1: **Empirical framework** + existing literature + existing data
 - ⇒ Estimates of losses across income distribution that are comparable across frictions
- Approach #2: Combine Approach #1 with a **life cycle model**
 - ✓ Accounts for dynamic effects and endogenous responses
 - ⇒ Estimates of effects on aggregate consumption and wealth inequality

- Individuals make many **choices** that are hard to justify with neoclassical models
 - Suboptimal refinancing, not claiming benefits, dominated insurance, ...
- Our questions:
 - ① What are the **financial impacts** of these “frictions”, individually and in aggregate?
 - ② How much do they contribute to **inequality** in income, consumption, and wealth?
- Approach #1: **Empirical framework** + existing literature + existing data
 - ⇒ Estimates of losses across income distribution that are comparable across frictions
- Approach #2: Combine Approach #1 with a **life cycle model**
 - ✓ Accounts for dynamic effects and endogenous responses
 - ⇒ Estimates of effects on aggregate consumption and wealth inequality

Takeaway: Effects of frictions are **large**, **heterogeneous**, and **regressive**

WHAT DO WE MEAN BY A “FRICTION”?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

WHAT DO WE MEAN BY A “FRICTION”?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

- 1 A friction is **not** necessarily a mistake

(“improvability”, Bernheim et al. 26)

WHAT DO WE MEAN BY A “FRICTION”?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

- 1 A friction is **not** necessarily a mistake (‘‘improvability’’, Bernheim et al. 26)
- 2 We **do not** specify the underlying microfoundation(s)

Scope: Estimate the **financial impact** of a large set of frictions, not

- Evaluate specific policies (Hendren-Sprung-Keyser 20)
- Measure the effects of specific biases (Stango-Zinman 23)
- Assess the welfare implications of removing frictions

WHAT DO WE MEAN BY A “FRICTION”?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

- ① A friction is **not** necessarily a mistake (“improvability”, Bernheim et al. 26)
- ② We **do not** specify the underlying microfoundation(s)

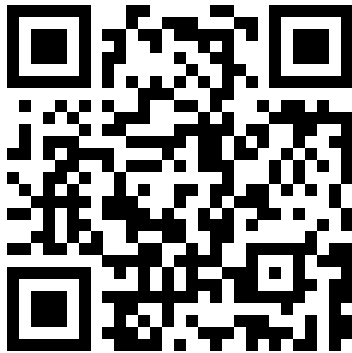
Scope: Estimate the **financial impact** of a large set of frictions, not

- Evaluate specific policies (Hendren-Sprung-Keyser 20)
- Measure the effects of specific biases (Stango-Zinman 23)
- Assess the welfare implications of removing frictions

- ③ If choice is not a friction, estimates are lower bound on utility costs to rationalize it

LIST OF FRICTIONS THAT WE CONSIDER

No.	Category	Friction	Syllabi Covering Friction		Coverage in Top Journals
			Top Universities (Out of 10)	High Schools (Out of 10)	
1	Education	Not Attending College Due to Frictions	0	10	.
2	Education	Attending 2-Year For-Profit Colleges	0	1	.
3	Savings	Not Meeting the Employer 401(k) Match	3	3	.
4	Savings	Not Investing in Stock Mutual Funds	10	10	.
5	Savings	Saving in Low Interest Accounts	5	4	.
6	Savings	Investing in Individual Stocks	6	7	.
7	Savings	Investing in High-Fee Mutual Funds	4	1	.
8	Borrowing	Not Refinancing a Mortgage	6	2	.
9	Borrowing	Not Getting an Additional Mortgage Quote	1	0	.
10	Borrowing	Suboptimally Repaying Credit Card Debt	0	0	.
11	Borrowing	Paying Late Fees	2	8	.
12	Borrowing	Paying Overdraft Fees	4	6	.
13	Borrowing	Not Filing for Bankruptcy Due to Frictions	3	2	.
14	Insurance	Choosing Dominated Health Insurance	0	2	.
15	Insurance	Suboptimally Lapsing Life Insurance	1	0	.
16	Benefits	Not Claiming EITC Benefits	1	1	.
17	Benefits	Not Claiming SNAP Benefits	0	0	.
18	Health	Not Taking Statin Medications	0	0	.
19	Health	Not Screening for Colon Cancer	0	0	.
20	Health	Failing to Quit Smoking	0	0	.



`timdesilva.me/frictions`

LIST OF FRICTIONS THAT WE CONSIDER

No.	Category	Friction	Syllabi Covering Friction		
			Top Universities (Out of 10)	High Schools (Out of 10)	Coverage in Top Journals
1	Education	Not Attending College Due to Frictions	0	10	.
2	Education	Attending 2-Year For-Profit Colleges	0	1	.
3	Savings	Not Meeting the Employer 401(k) Match	3	3	.
4	Savings	Not Investing in Stock Mutual Funds	10	10	.
5	Savings	Saving in Low Interest Accounts	5	4	.
6	Savings	Investing in Individual Stocks	6	7	.
7	Savings	Investing in High-Fee Mutual Funds	4	1	.
8	Borrowing	Not Refinancing a Mortgage	6	2	.
9	Borrowing	Not Getting an Additional Mortgage Quote	1	0	.
10	Borrowing	Suboptimally Repaying Credit Card Debt	0	0	.
11	Borrowing	Paying Late Fees	2	8	.
12	Borrowing	Paying Overdraft Fees	4	6	.
13	Borrowing	Not Filing for Bankruptcy Due to Frictions	3	2	.
14	Insurance	Choosing Dominated Health Insurance	0	2	.
15	Insurance	Suboptimally Lapsing Life Insurance	1	0	.
16	Benefits	Not Claiming EITC Benefits	1	1	.
17	Benefits	Not Claiming SNAP Benefits	0	0	.
18	Health	Not Taking Statin Medications	0	0	.
19	Health	Not Screening for Colon Cancer	0	0	.
20	Health	Failing to Quit Smoking	0	0	.

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

- 1 For some choices, we only consider a **subset** of them to be **frictions**
 - **Not attending college**: only measure non-attendance due to information frictions
 - **Not filing for bankruptcy**: only consider non-filing due to incorrect beliefs
 - **Failing to quit smoking**: only include smokers who report wanting to quit

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

- ① For some choices, we only consider a **subset** of them to be **frictions**
 - **Not attending college**: only measure non-attendance due to information frictions
 - **Not filing for bankruptcy**: only consider non-filing due to incorrect beliefs
 - **Failing to quit smoking**: only include smokers who report wanting to quit
- ② Notable examples of choices that we **do not** classify as frictions
 - **Gambling**: normative utility may include anticipatory or entertainment utility
 - **Using payday loans**: may be optimal if sufficiently credit-constrained
 - **Not building a credit score**: may not be possible given current circumstances

- 1 Empirical Framework and Examples (Approach #1)
- 2 Results (Approach #1)
- 3 Life Cycle Model (Sketch) and Results (Approach #2)
- 4 Conclusion

- 1 Empirical Framework and Examples (Approach #1)
- 2 Results (Approach #1)
- 3 Life Cycle Model (Sketch) and Results (Approach #2)
- 4 Conclusion

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low, Middle, High}\}$
 - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv 1$ (i at risk of friction at t), $f_{it} \equiv 1$ (i subject to friction at t)
 - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
 - f_{it} may depend on past choices (e.g. for-profit college attendance)

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low, Middle, High}\}$
 - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv \mathbf{1}$ (i at risk of friction at t), $f_{it} \equiv \mathbf{1}$ (i subject to friction at t)
 - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
 - f_{it} may depend on past choices (e.g. for-profit college attendance)

- Budget constraint:

$$\text{Consumption}_{it}(f_{it}) + \underbrace{\overbrace{\text{Wealth}_{it+1}(f_{it})}^{\text{carried out of } t} - \overbrace{\text{Wealth}_{it}(f_{it})}^{\text{carried into } t}}_{\text{net savings}} = \overbrace{R_{it}(f_{it})}^{\text{"consumable" income}}$$

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low, Middle, High}\}$
 - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv \mathbf{1}$ (i at risk of friction at t), $f_{it} \equiv \mathbf{1}$ (i subject to friction at t)
 - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
 - f_{it} may depend on past choices (e.g. for-profit college attendance)
- Budget constraint:

$$\text{Consumption}_{it}(f_{it}) + \text{Wealth}_{it+1}(f_{it}) - \text{Wealth}_{it}(f_{it}) = R_{it}(f_{it})$$

Definition of Financial Impact

$$l_{it}(f_{it}) \equiv R_{it}(0) - R_{it}(f_{it})$$

- **Individual** i observed at time t (**2010's**) in group $j_{it} \in \{\text{Low, Middle, High}\}$
 - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv \mathbf{1}$ (i at risk of friction at t), $f_{it} \equiv \mathbf{1}$ (i subject to friction at t)
 - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
 - f_{it} may depend on past choices (e.g. for-profit college attendance)
- Budget constraint:

$$\text{Consumption}_{it}(f_{it}) + \text{Wealth}_{it+1}(f_{it}) - \text{Wealth}_{it}(f_{it}) = R_{it}(f_{it})$$

Definition of Financial Impact

$$l_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Attending 2-Year For-Profit Colleges: Reduces Labor Income

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Not Investing in Stock Mutual Funds: Reduces **Capital Income**

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Not Claiming EITC Benefits: Reduces **Benefits**

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Paying Credit Card Late Fees: Creates **direct costs** from additional fees

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels

Failing to Quit Smoking: Creates **direct costs** from cigarettes and increases **mortality**

Definition of Financial Impact

$$I_{it}(f_{it}) \equiv R_{it}(0) A_{it}(0) - R_{it}(f_{it}) A_{it}(f_{it}), \quad A_{it}(f_{it}) = \mathbf{1} \text{ (} i \text{ alive at } t \text{)}$$

- Consumption and wealth not observed, so construct consumable income directly:

$$R_{it}(f_{it}) = \text{Labor Income}_{it}(f_{it}) + \text{Capital Income}_{it}(f_{it}) + \text{Benefits}_{it}(f_{it}) - \text{Taxes}_{it}(f_{it}) - \lambda_{it} f_{it}$$

- Different frictions create losses through different channels
- **Challenge:** fully accounting for interaction between channels + dynamic effects
 - Ignore dynamics in this analysis, and incorporate in **life cycle model**

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}(s_{it} = 1 \mid A_{it}(f_{it}) = 1, j_{it} = j)$

KEY PARAMETERS THAT WE ESTIMATE FOR EACH FRICTION

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}(s_{it} = 1 \mid A_{it}(f_{it}) = 1, j_{it} = j)$

Friction Prevalence: $F_j \equiv \mathbb{P}(f_{it} = 1 \mid s_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j)$

KEY PARAMETERS THAT WE ESTIMATE FOR EACH FRICTION

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}(s_{it} = 1 \mid A_{it}(f_{it}) = 1, j_{it} = j)$

Friction Prevalence: $F_j \equiv \mathbb{P}(f_{it} = 1 \mid s_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j)$

Average Loss: $L_j \equiv \frac{\mathbb{E}[f_{it} l_{it}(f_{it}) \mid j_{it} = j]}{\mathbb{E}[A_{it}(f_{it}) f_{it} \mid j_{it} = j]}$

KEY PARAMETERS THAT WE ESTIMATE FOR EACH FRICTION

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}(s_{it} = 1 \mid A_{it}(f_{it}) = 1, j_{it} = j)$

Friction Prevalence: $F_j \equiv \mathbb{P}(f_{it} = 1 \mid s_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j)$

Average Loss: $L_j \equiv \frac{\mathbb{E}[f_{it} l_{it}(f_{it}) \mid j_{it} = j]}{\mathbb{E}[A_{it}(f_{it}) f_{it} \mid j_{it} = j]}$

- Mortality **not** affected $\Rightarrow$ average loss simplifies to

$$L_j = \mathbb{E}[R_{it}(0) - R_{it}(f_{it}) \mid f_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j]$$

KEY PARAMETERS THAT WE ESTIMATE FOR EACH FRICTION

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}(s_{it} = 1 \mid A_{it}(f_{it}) = 1, j_{it} = j)$

Friction Prevalence: $F_j \equiv \mathbb{P}(f_{it} = 1 \mid s_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j)$

Average Loss: $L_j \equiv \frac{\mathbb{E}[f_{it} l_{it}(f_{it}) \mid j_{it} = j]}{\mathbb{E}[A_{it}(f_{it}) f_{it} \mid j_{it} = j]}$

- Mortality **not** affected $\Rightarrow$ average loss simplifies to

$$L_j = \mathbb{E}[R_{it}(0) - R_{it}(f_{it}) \mid f_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j]$$

- Mortality **is** affected $\Rightarrow$ who's alive changes, and average loss is

$$L_j = \frac{\text{total financial impact among } j}{\# \text{ of affected observations in } j \text{ that are currently alive}}$$

KEY PARAMETERS THAT WE ESTIMATE FOR EACH FRICTION

Definition of Key Parameters

Share at Risk: $S_j \equiv \mathbb{P}(s_{it} = 1 \mid A_{it}(f_{it}) = 1, j_{it} = j)$

Friction Prevalence: $F_j \equiv \mathbb{P}(f_{it} = 1 \mid s_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j)$

Average Loss: $L_j \equiv \frac{\mathbb{E}[f_{it} l_{it}(f_{it}) \mid j_{it} = j]}{\mathbb{E}[A_{it}(f_{it}) f_{it} \mid j_{it} = j]}$

Unconditional Loss: $\Delta_j \equiv \frac{\mathbb{E}[f_{it} l_{it}(f_{it}) \mid j_{it} = j]}{\mathbb{E}[A_{it}(f_{it}) \mid j_{it} = j]} = S_j F_j L_j$

- Mortality **not** affected $\Rightarrow$ average loss simplifies to

$$L_j = \mathbb{E}[R_{it}(0) - R_{it}(f_{it}) \mid f_{it} = 1, A_{it}(f_{it}) = 1, j_{it} = j]$$

- Mortality **is** affected $\Rightarrow$ who's alive changes, and average loss is

$$L_j = \frac{\text{total financial impact among } j}{\# \text{ of affected observations in } j \text{ that are currently alive}}$$

- Important parameters in life cycle model, but not needed for empirical approach:
 - **Correlation** of different frictions within-individual (in progress)
 - **Autocorrelation** of frictions within individual (for now: permanent)
 - **Sophistication** about frictions (for now: full sophistication)
- Caveats to both of our approaches:
 - Do not capture spillovers, GE effects, or **cross-subsidies** (could potentially incorporate)
 - Removing a particular (set of) friction(s) may not be **feasible** given microfoundations

EXAMPLES

- s_{it} = having a credit card, f_{it} = paying at least one late fee
 - Shares S_j and F_j measured by income group (CFPB Making Ends Meet)
 - Adjust for survey underreporting of credit cards (Karlan-Zinman 2008)
- Financial impact comes from λ_{it} = direct losses from paying late fees
 - No micro data on fees $\Rightarrow$ estimate one L_j for all j (CFPB CCM Report)
 - $L_j = (\text{average late fee}) \times (\# \text{ fees per card} \mid \text{fees} > 1) \times (\# \text{ cards per cardholder})$

EXAMPLE: PAYING LATE FEES

- s_{it} = having a credit card, f_{it} = paying at least one late fee
 - Shares S_j and F_j measured by income group (CFPB Making Ends Meet)
 - Adjust for survey underreporting of credit cards (Karlan-Zinman 2008)
- Financial impact comes from λ_{it} = direct losses from paying late fees
 - No micro data on fees $\Rightarrow$ estimate one L_j for all j (CFPB CCM Report)
 - $L_j = (\text{average late fee}) \times (\# \text{ fees per card} \mid \text{fees} > 1) \times (\# \text{ cards per cardholder})$

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.463 (0.013)	0.311 (0.017)	\$406 [381, 425]	\$57 [49, 64]
Middle Income	0.736 (0.004)	0.251 (0.010)	\$406 [381, 425]	\$73 [70, 78]
High Income	0.827 (0.007)	0.133 (0.011)	\$406 [381, 425]	\$44 [38, 50]

EXAMPLE: ATTENDING 2-YEAR FOR-PROFIT COLLEGES

- s_{it} = attending college, f_{it} = attending 2-year for-profit
 - Shares S_j and F_j measured by parental income group (BPS 12/17, Snyder et al. 2019)
 - Translate to child income group using intergenerational correlation (Haider-Solon 2006)
- Financial impact combines existing literature (Cellini-Turner 2019) + simulation
 - 1.5pp increase in unemployment probability $\Rightarrow \downarrow$ Labor Income
 - 11.3% earnings loss conditional on employment $\Rightarrow \downarrow$ Labor Income
 - \$6,100 increase in student debt (repaid under default plan) $\Rightarrow \lambda_{it}$

EXAMPLE: ATTENDING 2-YEAR FOR-PROFIT COLLEGES

- s_{it} = attending college, f_{it} = attending 2-year for-profit
 - Shares S_j and F_j measured by parental income group (BPS 12/17, Snyder et al. 2019)
 - Translate to child income group using intergenerational correlation (Haider-Solon 2006)
- Financial impact combines existing literature (Cellini-Turner 2019) + simulation
 - 1.5pp increase in unemployment probability $\Rightarrow \downarrow$ Labor Income
 - 11.3% earnings loss conditional on employment $\Rightarrow \downarrow$ Labor Income
 - \$6,100 increase in student debt (repaid under default plan) $\Rightarrow \lambda_{it}$

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.263 (0.016)	0.236 (0.014)	\$1,364 [1,116, 1,613]	\$85 [64, 106]
Middle Income	0.277 (0.010)	0.094 (0.007)	\$2,755 [1,961, 3,550]	\$72 [47, 97]
High Income	0.138 (0.012)	0.037 (0.011)	\$1,781 [447, 3,116]	\$9 [-14, 32]

EXAMPLE: NOT INVESTING IN STOCK MUTUAL FUNDS

- s_{it} = have financial wealth $\geq W_- \equiv \min\{2 \text{ mo. salary}, \$2K\}$ (SCF, Sabat-Gallagher 2019)
- f_{it} = not investing any wealth in diversified stocks (SCF)
 - Unlikely to be reflective of non-standard preferences (Choukhmane-de Silva 2026)
- Financial impact comes from $\downarrow$ Capital Income
 - $\ell_{it}(f_{it}) = \text{counterfactual equity share} \times (\text{financial wealth} - W_-)$
 - Counterfactual equity share = average age-specific equity share of those with $f_{it} = 0$

EXAMPLE: NOT INVESTING IN STOCK MUTUAL FUNDS

- s_{it} = have financial wealth $\geq W_- \equiv \min\{2 \text{ mo. salary}, \$2K\}$ (SCF, Sabat-Gallagher 2019)
- f_{it} = not investing any wealth in diversified stocks (SCF)
 - Unlikely to be reflective of non-standard preferences (Choukhmane-de Silva 2026)
- Financial impact comes from $\downarrow$ **Capital Income**
 - $\ell_{it}(f_{it}) = \text{counterfactual equity share} \times (\text{financial wealth} - W_-)$
 - Counterfactual equity share = average age-specific equity share of those with $f_{it} = 0$

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.326 (0.008)	0.646 (0.015)	\$354 [297, 436]	\$75 [61, 92]
Middle Income	0.643 (0.006)	0.305 (0.008)	\$959 [887, 1,039]	\$188 [171, 206]
High Income	0.867 (0.006)	0.129 (0.007)	\$2,615 [2,193, 3,076]	\$293 [239, 354]

EXAMPLE: FAILING TO QUIT SMOKING

- s_{it} = smoke before 50 and want to quit, f_{it} = want to quit before 50 but don't (NHIS)
- Choose age 50 as critical year for two reasons:
 - 1 Those who successfully quit before 50 have similar mortality to never-smokers
 - 2 Those who quit after 50 have similar mortality to non-smokers
- Financial impact comes from **mortality** and **direct losses**
 - Simulate increased ($\sim 2x$) mortality prob. by income-age $\Rightarrow A_{it}(0) - A_{it}(f_{it}) \geq 0$
 - Cost of cigarettes $\sim \$1.5K/\text{year} \Rightarrow \lambda_{it}$

EXAMPLE: FAILING TO QUIT SMOKING

- s_{it} = smoke before 50 and want to quit, f_{it} = want to quit before 50 but don't (NHIS)
- Choose age 50 as critical year for two reasons:
 - 1 Those who successfully quit before 50 have similar mortality to never-smokers
 - 2 Those who quit after 50 have similar mortality to non-smokers
- Financial impact comes from **mortality** and **direct losses**
 - Simulate increased ($\sim 2x$) mortality prob. by income-age $\Rightarrow A_{it}(0) - A_{it}(f_{it}) \geq 0$
 - Cost of cigarettes $\sim \$1.5K/\text{year} \Rightarrow \lambda_{it}$

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.211 (0.003)	0.624 (0.008)	\$8,707 [8,138, 10,263]	\$1,150 [1,074, 1,355]
Middle Income	0.240 (0.002)	0.465 (0.006)	\$11,259 [10,369, 13,507]	\$1,258 [1,158, 1,509]
High Income	0.222 (0.004)	0.353 (0.012)	\$19,375 [15,911, 45,530]	\$1,518 [1,247, 3,567]

EXAMPLE: FAILING TO QUIT SMOKING

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.211 (0.003)	0.624 (0.008)	\$8,707 [8,138, 10,263]	\$1,150 [1,074, 1,355]
Middle Income	0.240 (0.002)	0.465 (0.006)	\$11,259 [10,369, 13,507]	\$1,258 [1,158, 1,509]
High Income	0.222 (0.004)	0.353 (0.012)	\$19,375 [15,911, 45,530]	\$1,518 [1,247, 3,567]

EXAMPLE: FAILING TO QUIT SMOKING

Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.211 (0.003)	0.624 (0.008)	\$8,707 [8,138, 10,263]	\$1,150 [1,074, 1,355]
Middle Income	0.240 (0.002)	0.465 (0.006)	\$11,259 [10,369, 13,507]	\$1,258 [1,158, 1,509]
High Income	0.222 (0.004)	0.353 (0.012)	\$19,375 [15,911, 45,530]	\$1,518 [1,247, 3,567]

- Reduction in life-span ≈ 7 years; number of years subject to friction ≈ 22 years
- Average consumable income $\approx \$19K$; annual spend on cigarettes $\approx \$1.6K$

EXAMPLE: FAILING TO QUIT SMOKING

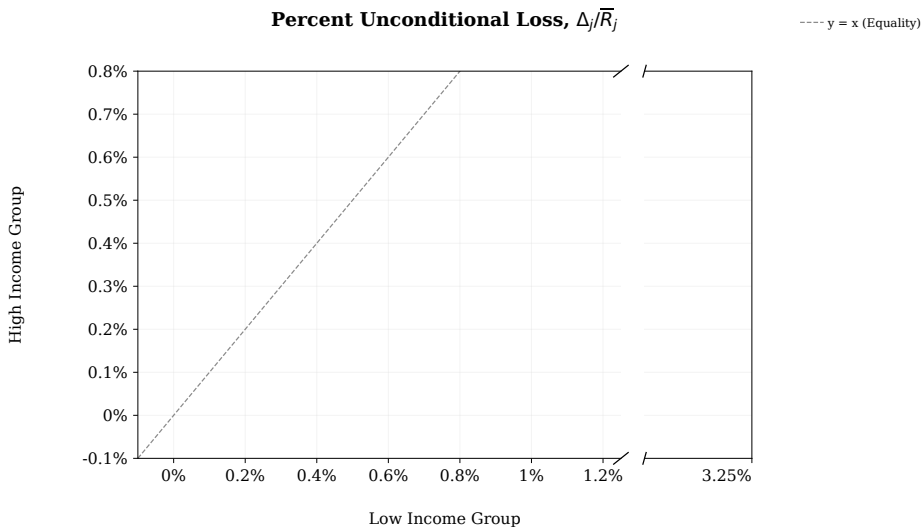
Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.211 (0.003)	0.624 (0.008)	\$8,707 [8,138, 10,263]	\$1,150 [1,074, 1,355]
Middle Income	0.240 (0.002)	0.465 (0.006)	\$11,259 [10,369, 13,507]	\$1,258 [1,158, 1,509]
High Income	0.222 (0.004)	0.353 (0.012)	\$19,375 [15,911, 45,530]	\$1,518 [1,247, 3,567]

- Reduction in life-span ≈ 7 years; number of years subject to friction ≈ 22 years
- Average consumable income $\approx \$19K$; annual spend on cigarettes $\approx \$1.6K$
- **Approximation:**

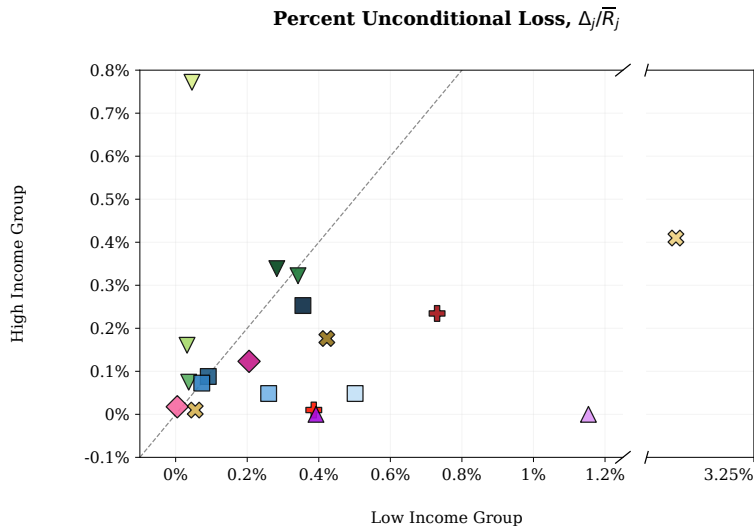
$$L_{\text{Low}} \approx \frac{\overbrace{7 \text{ years} \times \$19K}^{\text{lost income due to mortality}} + \overbrace{22 \text{ years} \times \$1.6K}^{\text{direct loss from cigarettes}}}{22 \text{ years}} \approx \text{\textcolor{brown}{\$8K / year}}$$

- 1 Empirical Framework and Examples (Approach #1)
- 2 Results (Approach #1)
- 3 Life Cycle Model (Sketch) and Results (Approach #2)
- 4 Conclusion

FINANCIAL IMPACTS AS % OF PERMANENT CONSUMABLE INCOME



FINANCIAL IMPACTS AS % OF PERMANENT CONSUMABLE INCOME



---- y = x (Equality)

Education

- ✚ Not Attending College Due to Frictions
- ✚ Attending 2-Year For-Profit Colleges

Savings

- ▼ Not Meeting the Employer 401(k) Match
- ▼ Not Investing in Stock Mutual Funds
- ▼ Saving in Low Interest Accounts
- ▼ Investing in Individual Stocks
- ▼ Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- ◆ Choosing Dominated Health Insurance
- ◆ Suboptimally Lapsing Life Insurance

Benefits

- ▲ Not Claiming EITC Benefits
- ▲ Not Claiming SNAP Benefits

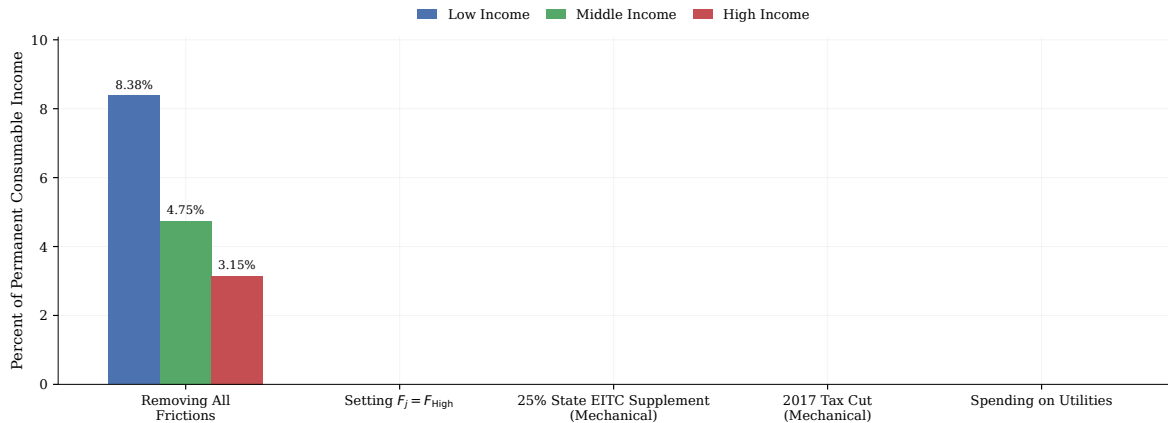
Health

- ✕ Not Taking Statin Medications
- ✕ Not Screening for Colon Cancer
- ✕ Not Quitting Smoking

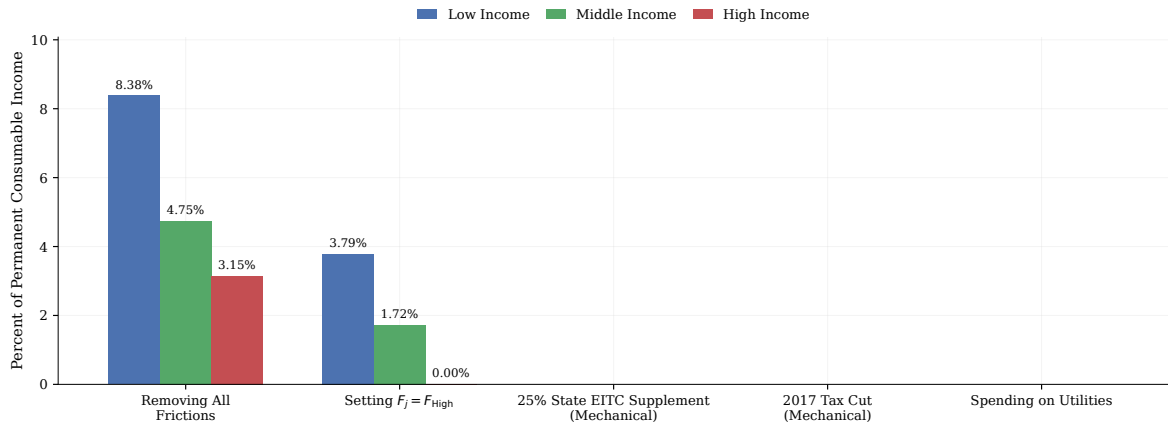
Income Group	Average	All
Low	0.44%	8.38%
High	0.17%	3.15%

► Dollars

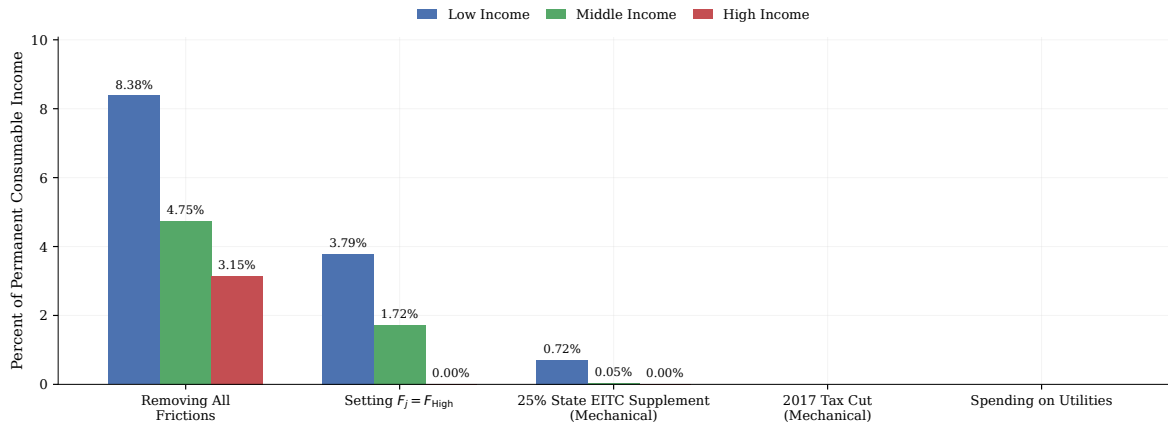
BENCHMARKING TOTAL LOSSES FROM FRICTIONS



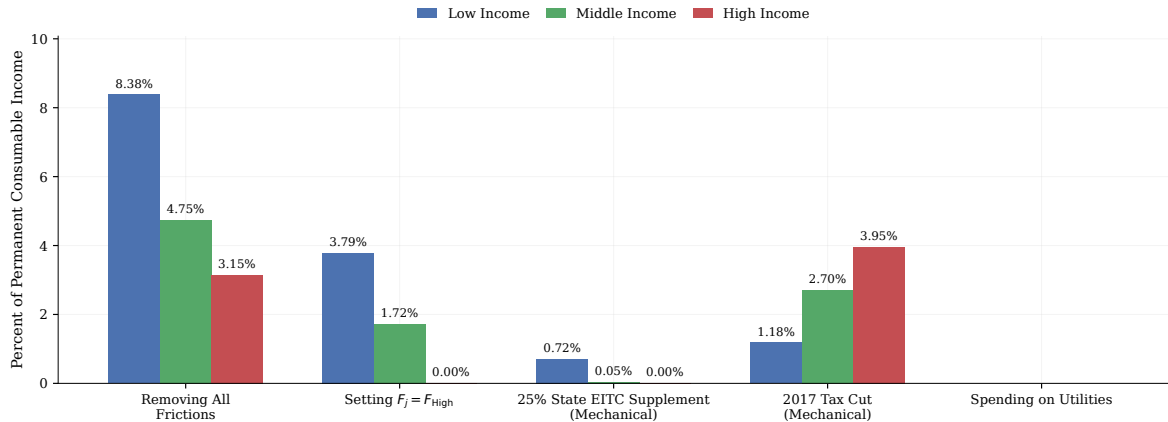
BENCHMARKING TOTAL LOSSES FROM FRICTIONS



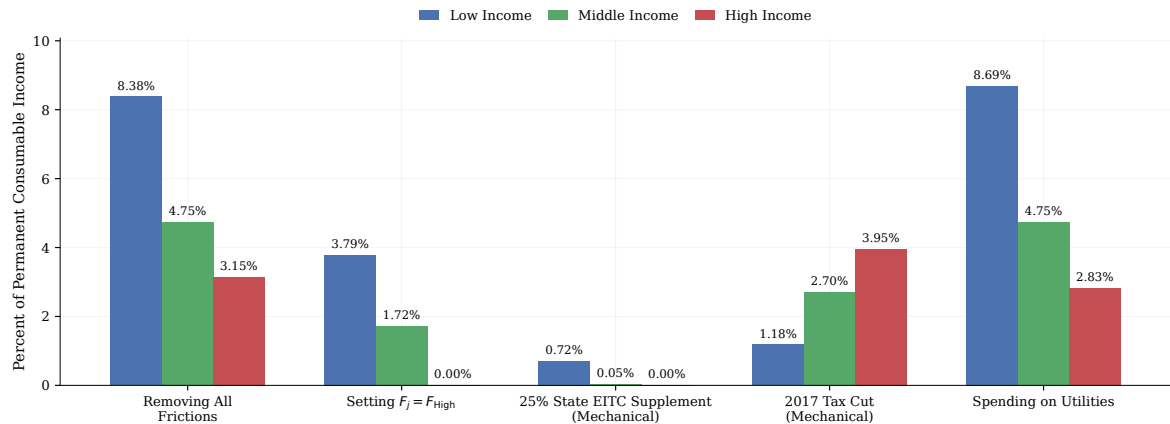
BENCHMARKING TOTAL LOSSES FROM FRICTIONS



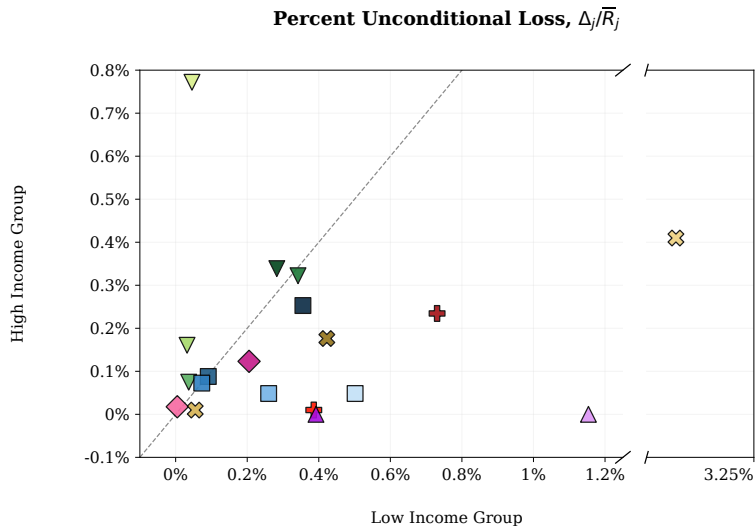
BENCHMARKING TOTAL LOSSES FROM FRICTIONS



BENCHMARKING TOTAL LOSSES FROM FRICTIONS



FINANCIAL IMPACTS: CATEGORY AVERAGES



---- y = x (Equality)

Education

- ✚ Not Attending College Due to Frictions
- ✚ Attending 2-Year For-Profit Colleges

Savings

- ▼ Not Meeting the Employer 401(k) Match
- ▼ Not Investing in Stock Mutual Funds
- ▼ Saving in Low Interest Accounts
- ▼ Investing in Individual Stocks
- ▼ Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- ◆ Choosing Dominated Health Insurance
- ◆ Suboptimally Lapsing Life Insurance

Benefits

- ▲ Not Claiming EITC Benefits
- ▲ Not Claiming SNAP Benefits

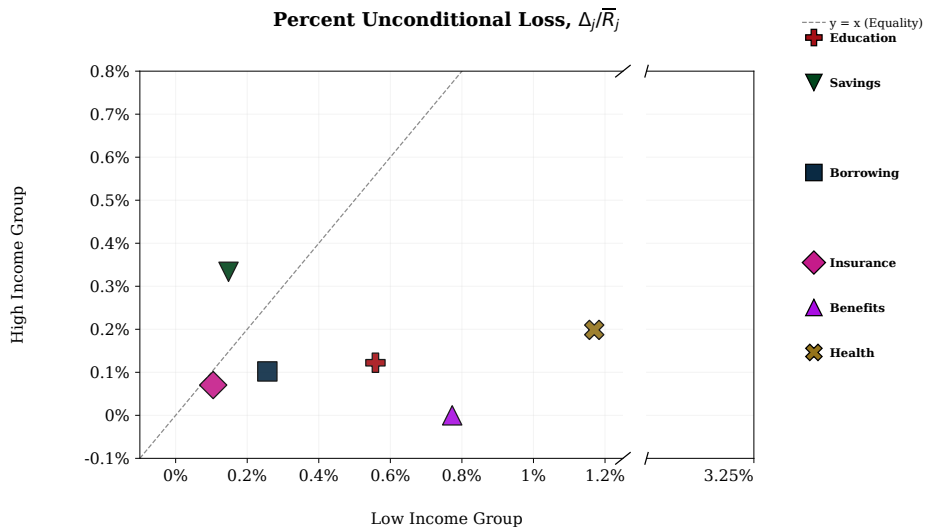
Health

- ✚ Not Taking Statin Medications
- ✚ Not Screening for Colon Cancer
- ✚ Not Quitting Smoking

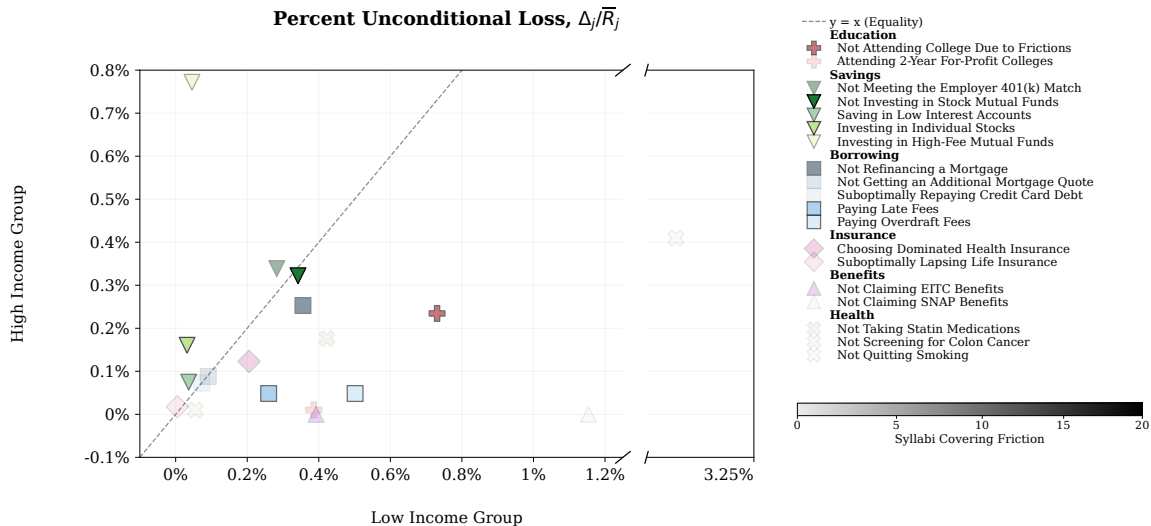
Income Group	Average	All
Low	0.44%	8.38%
High	0.17%	3.15%

► Dollars

FINANCIAL IMPACTS: CATEGORY AVERAGES

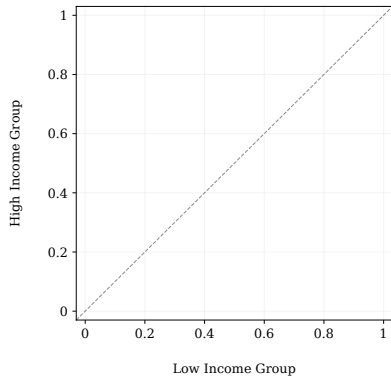


FINANCIAL IMPACT OF FRICTIONS: SYLLABUS COVERAGE

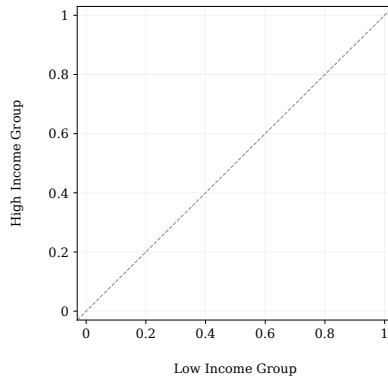


DECOMPOSING FINANCIAL IMPACT: SHARES AND PREVALENCE

Share at Risk, S_j



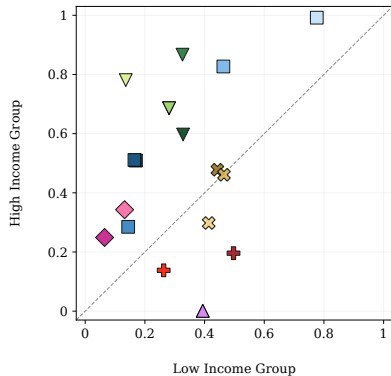
Friction Prevalence, F_j



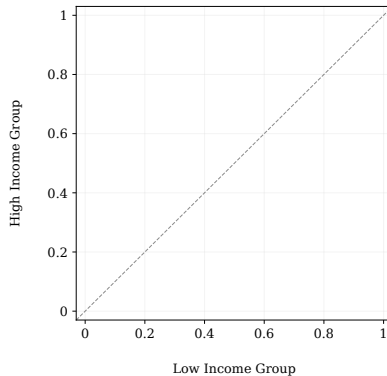
---- $y = x$ (Equality)

DECOMPOSING FINANCIAL IMPACT: SHARES AND PREVALENCE

Share at Risk, S_j



Friction Prevalence, F_j



---- $y = x$ (Equality)

Education

- Not Attending College Due to Frictions
- Attending 2-Year For-Profit Colleges

Savings

- Not Meeting the Employer 401(k) Match
- Not Investing in Stock Mutual Funds
- Saving in Low Interest Accounts
- Investing in Individual Stocks
- Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- Choosing Dominated Health Insurance
- Suboptimally Lapsing Life Insurance

Benefits

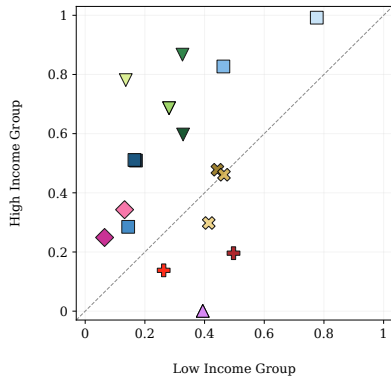
- Not Claiming EITC Benefits
- Not Claiming SNAP Benefits

Health

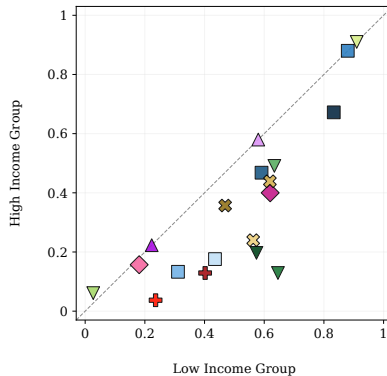
- Not Taking Statin Medications
- Not Screening for Colon Cancer
- Not Quitting Smoking

DECOMPOSING FINANCIAL IMPACT: SHARES AND PREVALENCE

Share at Risk, S_j



Friction Prevalence, F_j



---- y = x (Equality)

Education

- Not Attending College Due to Frictions
- Attending 2-Year For-Profit Colleges

Savings

- Not Meeting the Employer 401(k) Match
- Not Investing in Stock Mutual Funds
- Saving in Low Interest Accounts
- Investing in Individual Stocks
- Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- Choosing Dominated Health Insurance
- Suboptimally Lapsing Life Insurance

Benefits

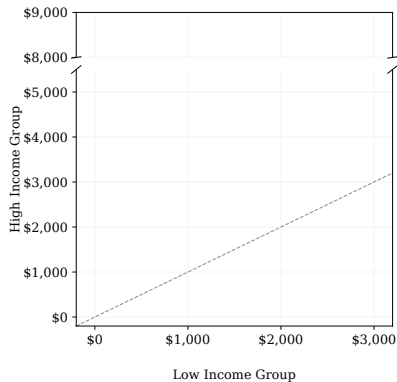
- Not Claiming EITC Benefits
- Not Claiming SNAP Benefits

Health

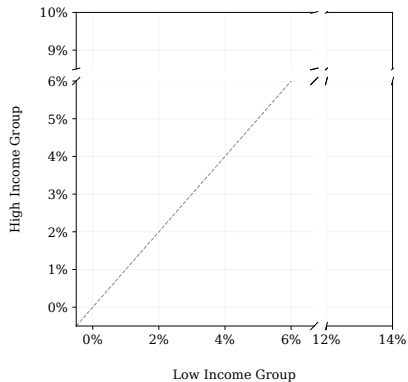
- Not Taking Statin Medications
- Not Screening for Colon Cancer
- Not Quitting Smoking

DECOMPOSING FINANCIAL IMPACT: AVERAGE LOSSES

Average Loss, L_j

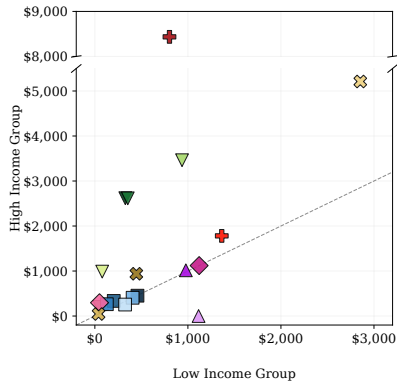


Percent Average Loss, $L_j/\bar{R}_j$

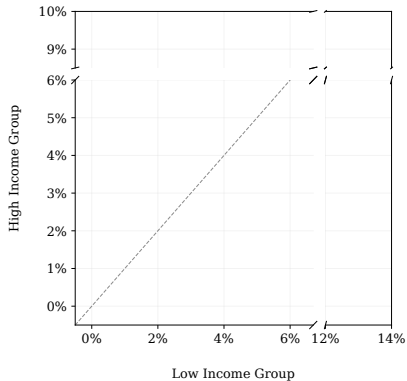


DECOMPOSING FINANCIAL IMPACT: AVERAGE LOSSES

Average Loss, L_j



Percent Average Loss, $L_j/\bar{R}_j$



---- $y = x$ (Equality)

Education

- Not Attending College Due to Frictions
- Attending 2-Year For-Profit Colleges

Savings

- Not Meeting the Employer 401(k) Match
- Not Investing in Stock Mutual Funds
- Saving in Low Interest Accounts
- Investing in Individual Stocks
- Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- Choosing Dominated Health Insurance
- Suboptimally Lapsing Life Insurance

Benefits

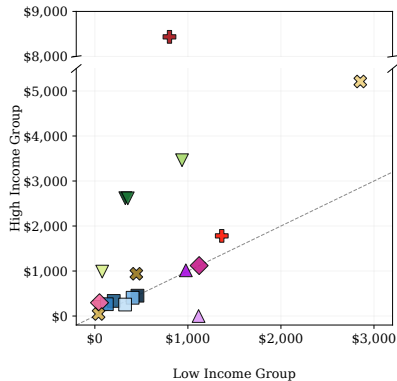
- Not Claiming EITC Benefits
- Not Claiming SNAP Benefits

Health

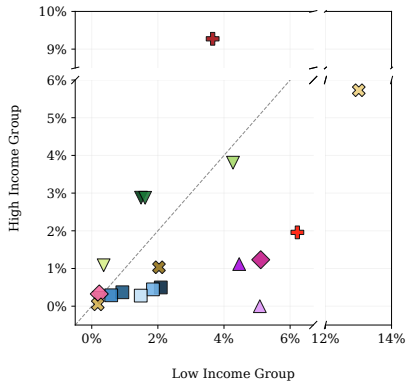
- Not Taking Statin Medications
- Not Screening for Colon Cancer
- Not Quitting Smoking

DECOMPOSING FINANCIAL IMPACT: AVERAGE LOSSES

Average Loss, L_j



Percent Average Loss, $L_j/\bar{R}_j$



----- $y = x$ (Equality)

Education

- Not Attending College Due to Frictions
- Attending 2-Year For-Profit Colleges

Savings

- Not Meeting the Employer 401(k) Match
- Not Investing in Stock Mutual Funds
- Saving in Low Interest Accounts
- Investing in Individual Stocks
- Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- Choosing Dominated Health Insurance
- Suboptimally Lapsing Life Insurance

Benefits

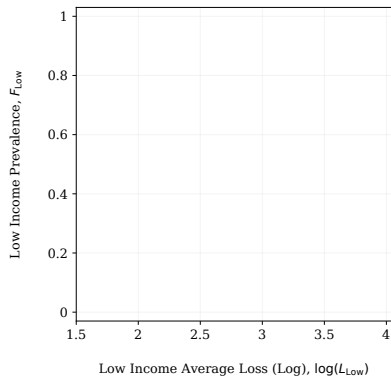
- Not Claiming EITC Benefits
- Not Claiming SNAP Benefits

Health

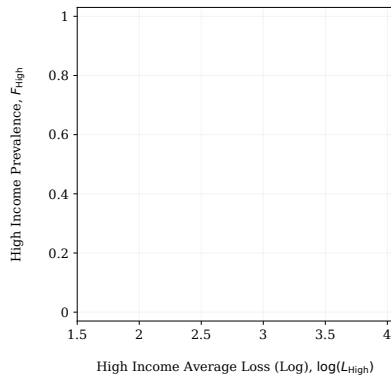
- Not Taking Statin Medications
- Not Screening for Colon Cancer
- Not Quitting Smoking

RELATIONSHIP BETWEEN PREVALENCE AND AVERAGE LOSS

Low Income Group

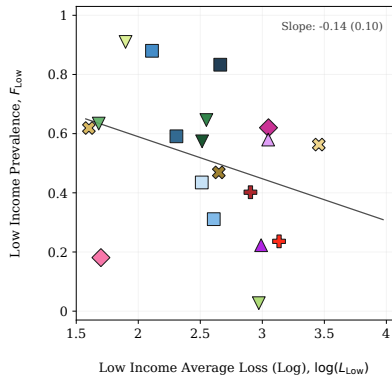


High Income Group

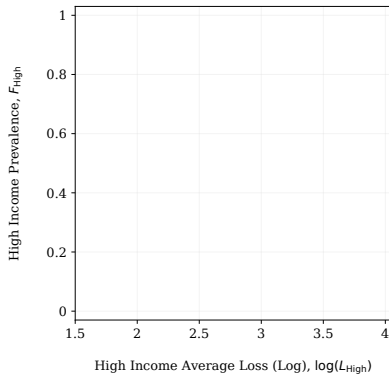


RELATIONSHIP BETWEEN PREVALENCE AND AVERAGE LOSS

Low Income Group



High Income Group



Education

- Not Attending College Due to Frictions
- Attending 2-Year For-Profit Colleges

Savings

- Not Meeting the Employer 401(k) Match
- Not Investing in Stock Mutual Funds
- Saving in Low Interest Accounts
- Investing in Individual Stocks
- Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- Choosing Dominated Health Insurance
- Suboptimally Lapsing Life Insurance

Benefits

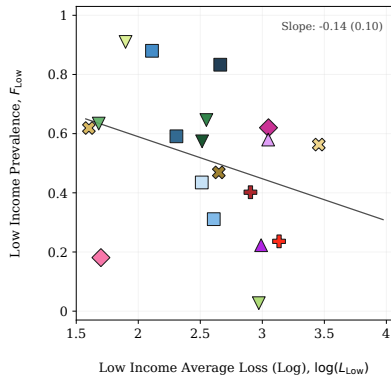
- Not Claiming EITC Benefits
- Not Claiming SNAP Benefits

Health

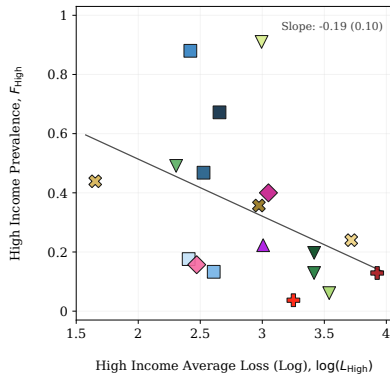
- Not Taking Statin Medications
- Not Screening for Colon Cancer
- Not Quitting Smoking

RELATIONSHIP BETWEEN PREVALENCE AND AVERAGE LOSS

Low Income Group



High Income Group



Education

- ✚ Not Attending College Due to Frictions
- ✚ Attending 2-Year For-Profit Colleges

Savings

- ▼ Not Meeting the Employer 401(k) Match
- ▼ Not Investing in Stock Mutual Funds
- ▼ Saving in Low Interest Accounts
- ▼ Investing in Individual Stocks
- ▼ Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- ◆ Choosing Dominated Health Insurance
- ◆ Suboptimally Lapsing Life Insurance

Benefits

- ▲ Not Claiming EITC Benefits
- ▲ Not Claiming SNAP Benefits

Health

- ✕ Not Taking Statin Medications
- ✕ Not Screening for Colon Cancer
- ✕ Not Quitting Smoking

- ① **Evolution** of key parameters over time
 - Losses seem to respond to policies (e.g. for-profit certifications, public health efforts)
- ② **Correlation** of prevalence across-frictions within-individual
 - Currently have estimated 6 bivariate correlations with mean 0.032 (SE 0.006)
- ③ Comparison of losses with **forecasts** from a survey of academics
- ④ Comparison of losses with coverage in **academic journals** and **media**

- ① **Evolution** of key parameters over time
 - Losses seem to respond to policies (e.g. for-profit certifications, public health efforts)
 - ② **Correlation** of prevalence across-frictions within-individual
 - Currently have estimated 6 bivariate correlations with mean 0.032 (SE 0.006)
 - ③ Comparison of losses with **forecasts** from a survey of academics
 - ④ Comparison of losses with coverage in **academic journals** and **media**
- ... More ideas welcome!

- 1 Empirical Framework and Examples (Approach #1)
- 2 Results (Approach #1)
- 3 Life Cycle Model (Sketch) and Results (Approach #2)
- 4 Conclusion

- Reasons to develop a life cycle model:
 - ① Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
 - ② Removing a friction likely creates **endogenous** responses
 - ③ Study affects on aggregate **consumption** and **wealth** inequality

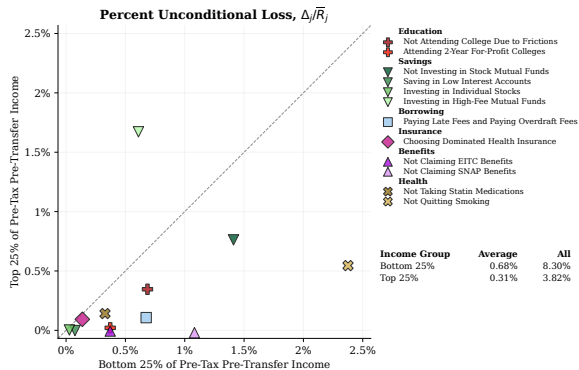
- Reasons to develop a life cycle model:
 - 1 Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
 - 2 Removing a friction likely creates **endogenous** responses
 - 3 Study affects on aggregate **consumption** and **wealth** inequality
- Key model ingredients (partial equilibrium):
 - Individuals ex-ante heterogeneous in permanent income and education
 - Individuals make dynamic consumption-savings and portfolio choice decisions
 - Available assets: liquid savings, 401(k), credit cards, housing
 - Non-Gaussian income risk, mortality risk, realistic social insurance system

- Reasons to develop a life cycle model:
 - ① Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
 - ② Removing a friction likely creates **endogenous** responses
 - ③ Study affects on aggregate **consumption** and **wealth** inequality
- Key model ingredients (partial equilibrium):
 - Individuals ex-ante heterogeneous in permanent income and education
 - Individuals make dynamic consumption-savings and portfolio choice decisions
 - Available assets: liquid savings, 401(k), credit cards, housing
 - Non-Gaussian income risk, mortality risk, realistic social insurance system
- Individuals subject to frictions $\mathbf{f}_i \in \{0, 1\}^{20}$ based on **empirical prevalence**
 - Permanent throughout life, full sophistication, no correlation (for now)

- Reasons to develop a life cycle model:
 - ① Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
 - ② Removing a friction likely creates **endogenous** responses
 - ③ Study affects on aggregate **consumption** and **wealth** inequality
- Key model ingredients (partial equilibrium):
 - Individuals ex-ante heterogeneous in permanent income and education
 - Individuals make dynamic consumption-savings and portfolio choice decisions
 - Available assets: liquid savings, 401(k), credit cards, housing
 - Non-Gaussian income risk, mortality risk, realistic social insurance system
- Individuals subject to frictions $\mathbf{f}_i \in \{0, 1\}^{20}$ based on **empirical prevalence**
 - Permanent throughout life, full sophistication, no correlation (for now)
- **Key counterfactuals:** Set $\mathbf{f}_i(n) = 0$ for each friction n and compare with data

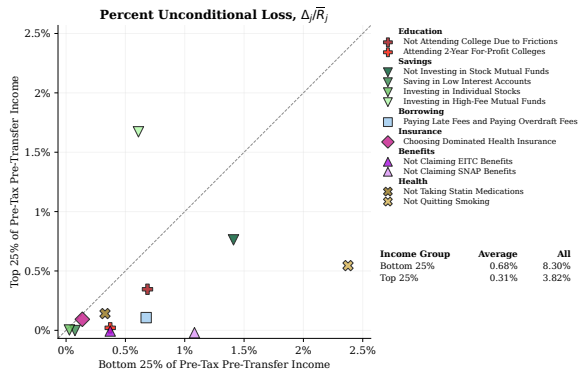
FINANCIAL IMPACT OF FRICTIONS: TWO APPROACHES

Life Cycle Model

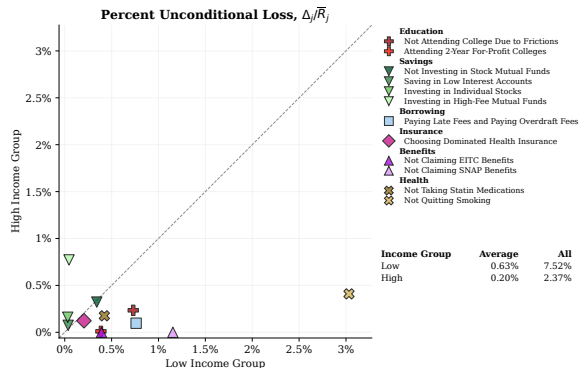


FINANCIAL IMPACT OF FRICTIONS: TWO APPROACHES

Life Cycle Model

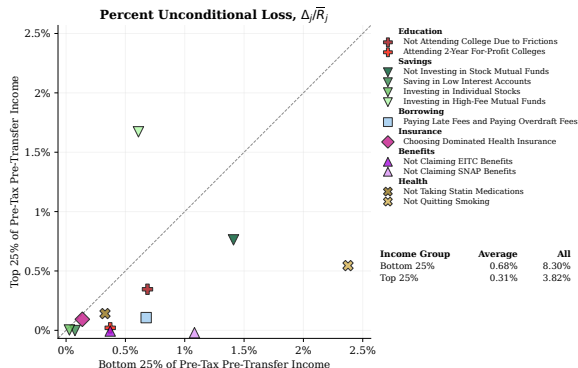


Empirical Estimates

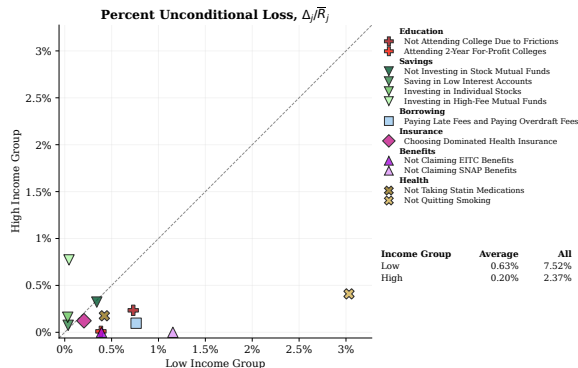


FINANCIAL IMPACT OF FRICTIONS: TWO APPROACHES

Life Cycle Model

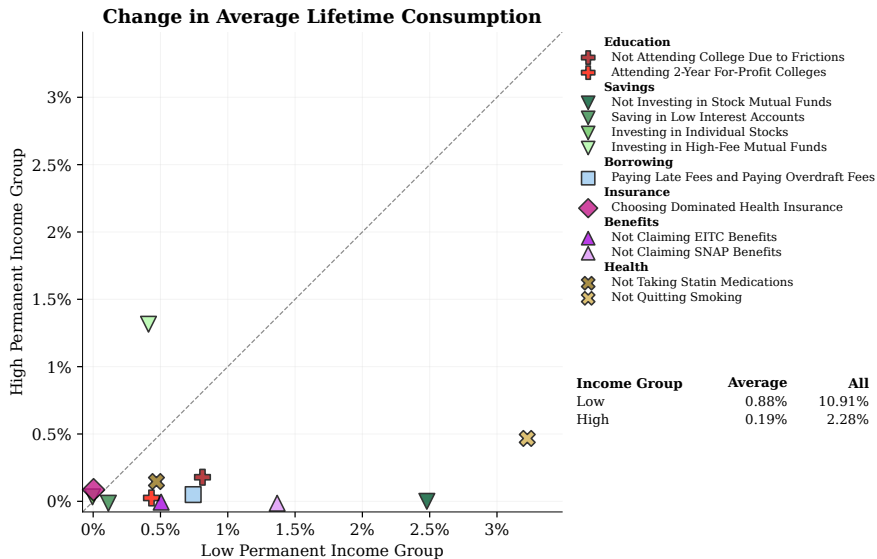


Empirical Estimates

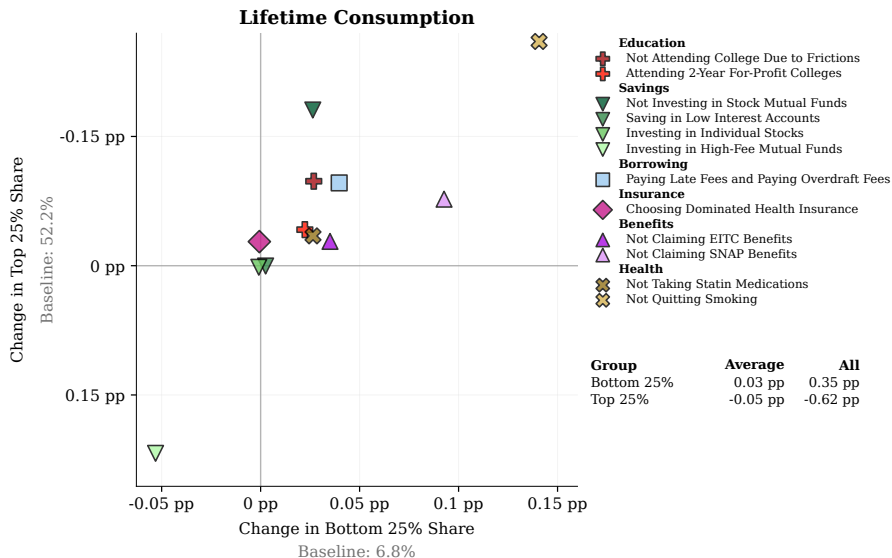


Losses are much larger for **savings** frictions, due to **dynamic** effects

FRICTIONS HAVE LARGE EFFECTS ON LIFETIME CONSUMPTION



FRICTIONS REDUCE LIFETIME CONSUMPTION INEQUALITY



- 1 Empirical Framework and Examples (Approach #1)
- 2 Results (Approach #1)
- 3 Life Cycle Model (Sketch) and Results (Approach #2)
- 4 Conclusion

- ① Decision-making frictions create meaningful losses in aggregate
 - Aggregate losses are **8%** (**3%**) for low (high) income = expenditure on utilities
 - Largest (as of now): smoking, fund fees, college attendance, SNAP benefits
- ② Losses from decision-making frictions are **regressive**
 - High-income have more to lose (L), but low-income **prevalence** (F) is higher
 - Main exception: Savings frictions, which are most covered by syllabi
 - **Model:** Removing frictions decreases inequality in lifetime consumption

- ① Decision-making frictions create meaningful losses in aggregate
 - Aggregate losses are **8%** (**3%**) for low (high) income = expenditure on utilities
 - Largest (as of now): smoking, fund fees, college attendance, SNAP benefits
- ② Losses from decision-making frictions are **regressive**
 - High-income have more to lose (L), but low-income **prevalence** (F) is higher
 - Main exception: Savings frictions, which are most covered by syllabi
 - **Model**: Removing frictions decreases inequality in lifetime consumption

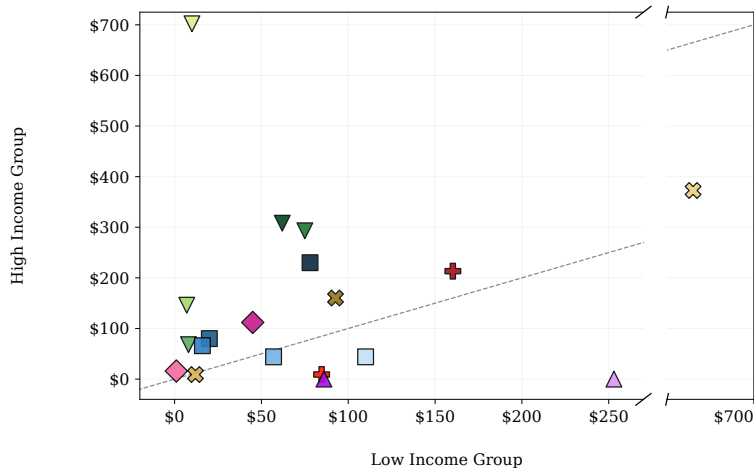
... Much more to come!

THANK YOU!

`sdellavi@berkeley.edu`
`tdesilva@stanford.edu`
`rohanjha@g.harvard.edu`

FINANCIAL IMPACTS IN DOLLARS

Unconditional Loss, Δ_j



---- $y = x$ (Equality)

Education

- Not Attending College Due to Frictions
- Attending 2-Year For-Profit Colleges

Savings

- Not Meeting the Employer 401(k) Match
- Not Investing in Stock Mutual Funds
- Saving in Low Interest Accounts
- Investing in Individual Stocks
- Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- Choosing Dominated Health Insurance
- Suboptimally Lapsing Life Insurance

Benefits

- Not Claiming EITC Benefits
- Not Claiming SNAP Benefits

Health

- Not Taking Statin Medications
- Not Screening for Colon Cancer
- Not Quitting Smoking

Income Group	Average	All
Low	\$97	\$1,839
High	\$151	\$2,862

◀ Back

FINANCIAL IMPACTS: CATEGORY AVERAGES IN DOLLARS

